Actual Miller computations and quotient-domain disproofs

For E/F5: $y^2=x^3+1$ and the actual field $L=F25=F5[\beta]$, $\beta^2=2$, Lean verifies both 9×9 order-three Miller tables and every used nonzero evaluation. It proves bilinearity and nondegeneracy on the actual finite quotient domains, plus direct base-field domain counterexamples. [pcc:unverified]

The functions are prescribed Miller expressions with actual curve/field operations. Their canonical divisor-defined pairing identification remains a separate formal obligation. Tangent triple-intersection certificates are included; no divisor/valuation theorem is assumed. [pcc:unverified]

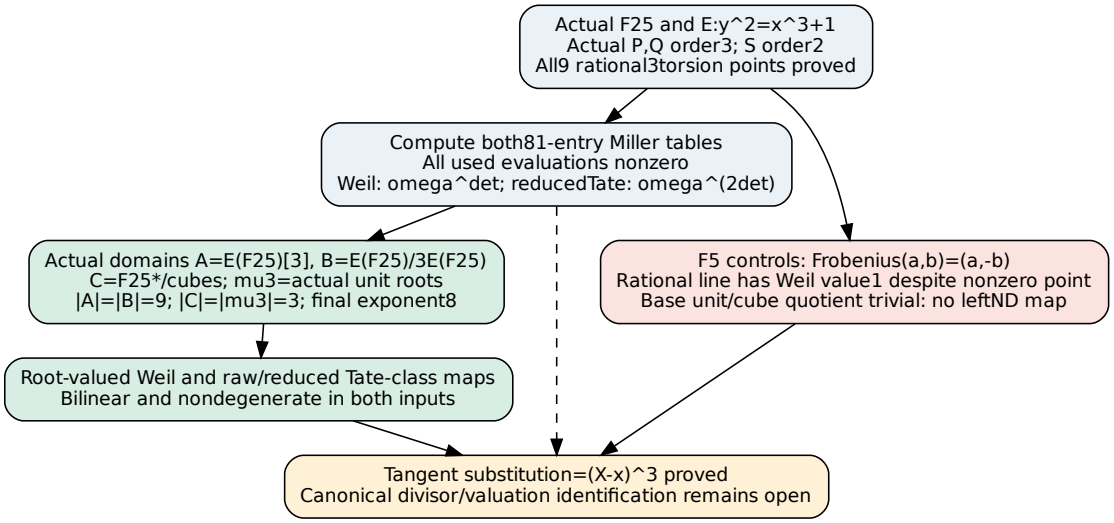


Figure 1: Actual computations, quotient domains, counterexamples and geometric boundary. [pcc:unverified]

Concrete formulas and actual proof map

P=(0,1), Q=(1,β), S=(4,0), ω=2+3β. The actual subgroup A=E(L)[3] has nine points, indexed by aP+bQ with a,b in Z/3. The index map uses actual Mathlib point addition; injectivity and exhaustive torsion coverage are proved. P/Q have order3; S is nonzero order2. [pcc:unverified]

For nonzero U=(x,y), use f_U(R)=y_R−y−(3x²/(2y))(x_R−x), with f_O=1. The checked expressions are W(U,V)=f_V(U+S)/f_V(S) · f_U(−S)/f_U(V−S) and T(U,V)=(f_U(V+S)/f_U(S))⁸. All used values are nonzero. Every ordered pair of indexed torsion points satisfies W=ω^(ad−bc), T=ω^(2(ad−bc)). In particular W(P,Q)=ω and T(P,Q)=ω². Orientation follows Galbraith(26.1). [pcc:unverified]

Module	Exact obligation
MillerCurve	Actual field, primitive root, curve and P/Q/S orders
TorsionIndex	Actual index homomorphism, injectivity, exhaustive coverage/cardinality9
MillerExpressions	Formulas, generator values, auxiliary support separation
MillerTable	Both full81-entry tables and nonzero evaluations
MillerProperties	Finite-expression bilinearity/nondegeneracy and cyclic restriction failure
BaseDescent	Actual Frobenius/base image, Q nonrational, base target trivial and direct negations
TateQuotient	Every actual point killed by6; actual quotient representatives/cardinality9
UnitsTarget	Final-exponent kernel/surjectivity, target isomorphism and cardinalities3
TateClassPairing	Raw/reduced maps on actual quotient domains and their group properties
WeilTarget	Root-valued alternating Weil expression and base restriction disproof
TangentCertificate	Triple-intersection polynomial identity, actual coefficient checks and universal ring proof
MillerCertificate	Combined audit theorem and exact generator roots

Every actual E(L) point is killed by6. The actual inclusion A→E(L)/3E(L) is bijective; each class has its unique 3-torsion representative, equivalently 4R for a point representative R. The raw Miller ratio therefore defines an actual unit class in C=L/(L)³ on the correct right quotient. Final exponent8 has kernel exactly the cubes and induces C≃μ3(L). Both raw and reduced maps are bilinear and nondegenerate in both variables. The Weil expression is also alternating and bilaterally nondegenerate. [pcc:unverified]

Skeptical controls, provenance and exact assurance

Actual Frobenius sends (a,b) to (a,−b). The F5-rational image in A is exactly the cyclic P-line, while Q lies outside it. W restricted to that line is identically1 despite a nonzero input; Lean directly negates the resulting nondegeneracy assertion. The actual base target F5*/(F5*)³ is trivial because cubing is surjective. An actual nonzero order3 base point therefore precludes every left-nondegenerate map to that target. The checks are at extension degrees1/2; gcd(3,5)=1, 3 not dividing4, and3 dividing24 are proved. Correct root-of-unity hypotheses remain intact. [pcc:unverified]

The tangent polynomial substitution is exactly $(X-x)^3$ at every nonzero actual 3-torsion point, with nonzero y. Finite field coefficient identities and a universal polynomial ring proof establish it. A projective local-chart factor and a nonzero numerator at infinity are also checked. Turning these identities into a complete principal-divisor/valuation identification remains open. The packet does not import a source nondegeneracy theorem as an axiom. [pcc:unverified]

Full project gate:126declarations, zero failures and zero trusted-native exceptions. All51attempt/source files and five whole-gate files have byte-equal archive roundtrip checks. Thirty-two formal/config dependency files are pinned in sources.json. Lean4.35.0 and Mathlib56a00585; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, waived linters or increased elaboration limits. New source imports serialize heavy branches for local memory; issued parents are unchanged. Actual local replay and independent signed standing are recorded separately in verification.json. Signed standing remains unverified. [pcc:unverified]

Primary reading: Galbraith26.2 alternative Weil definition(26.1), 26.3 Tate(26.2), MillerAlgorithm29 and reduced Tate26.3.2, including support, quotient, orientation and field hypotheses. Frey2009 selected 3.1.1–3.4/5.7–5.10 has different local/cohomological domains and is not substituted for the finite-field construction. Original Frey–Rück1994 remains unread: correct DOI suffix−6 is confirmed by Crossref, but the AMS PDF still returns403; old sealed suffix−3 remains historical. Exact hashes, URLs, read scopes and metadata receipts are retained. [pcc:unverified]

The finite checks do not prove a generic all-curve nondegeneracy, geometric torsion-rank, MOV/Frey–Rück reduction or complexity theorem. Base-domain controls do not refute correctly qualified source results. No native experiment, sampling, timing, benchmark, independent-host or paid-compute claim. KB MCP absent/wrapper127; bounded primary/repository/Mathlib fallback. Proof map, exact statement and dependencies: README.md and sources.json. Actual packet/verdict/standing: verification.json. [pcc:unverified]